NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-22885/92A(EC)
10 December 1981
SUPERSEDING
MIL-S-22885/92(EC)
21 June 1979

PERFORMANCE SPECIFICATION SHEET

SWITCH, PUSHBUTTON, ILLUMINATED, ROCKER, PADDLE, POWER DUTY, UNSEALED

This specification sheet is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this document and the latest issue of MIL-S-22885.

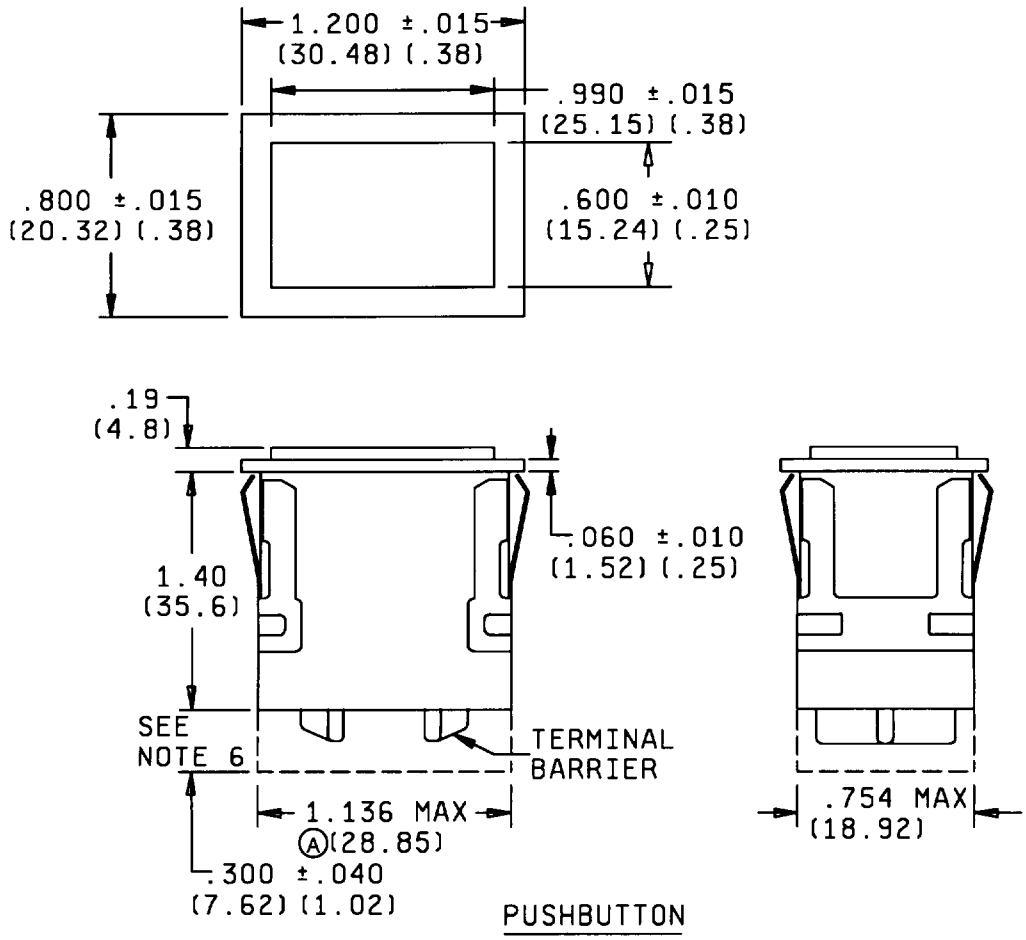
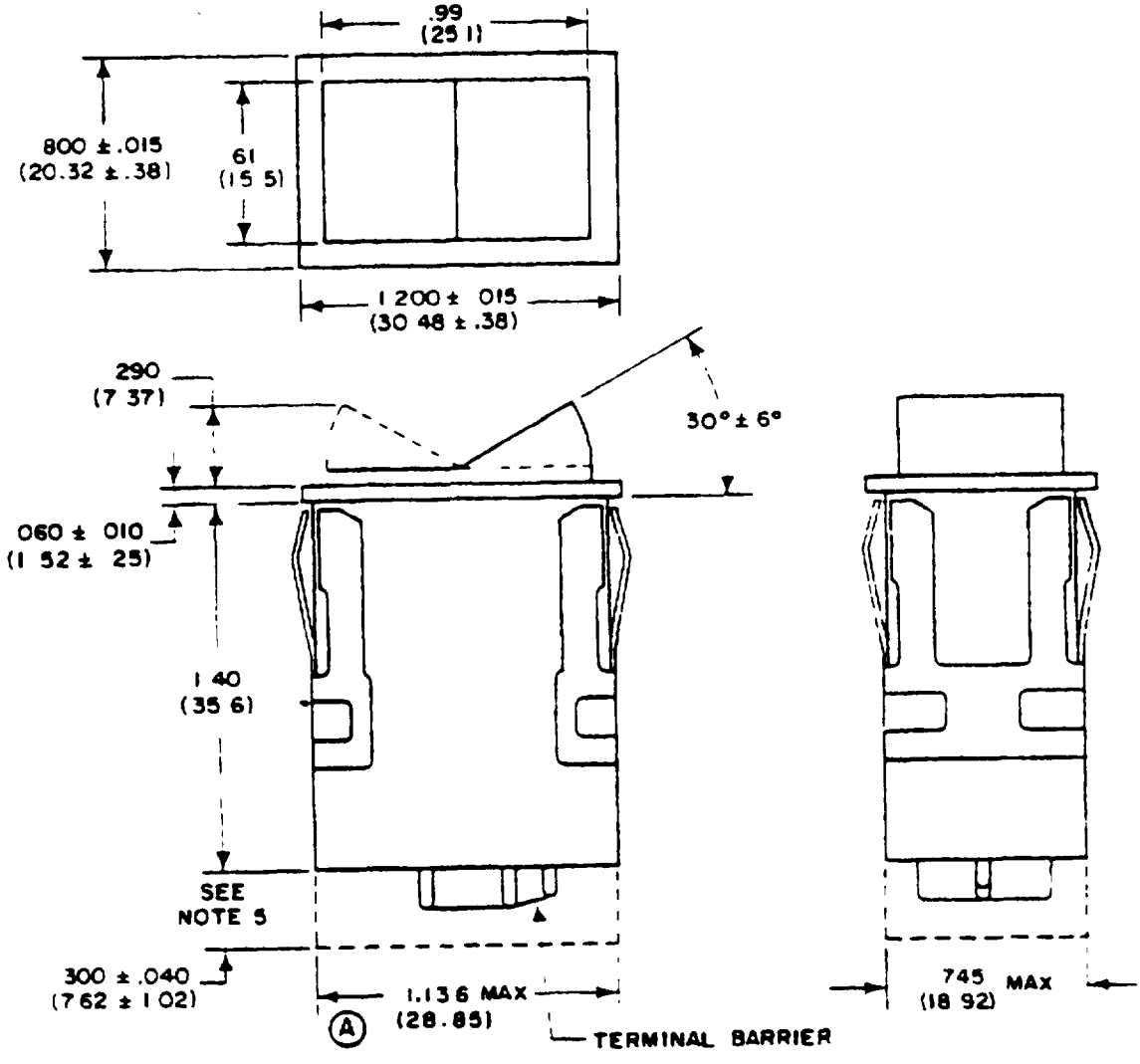


FIGURE 1. Configurations and dimensions .

Ⓐ denotes changes

FSC 5930

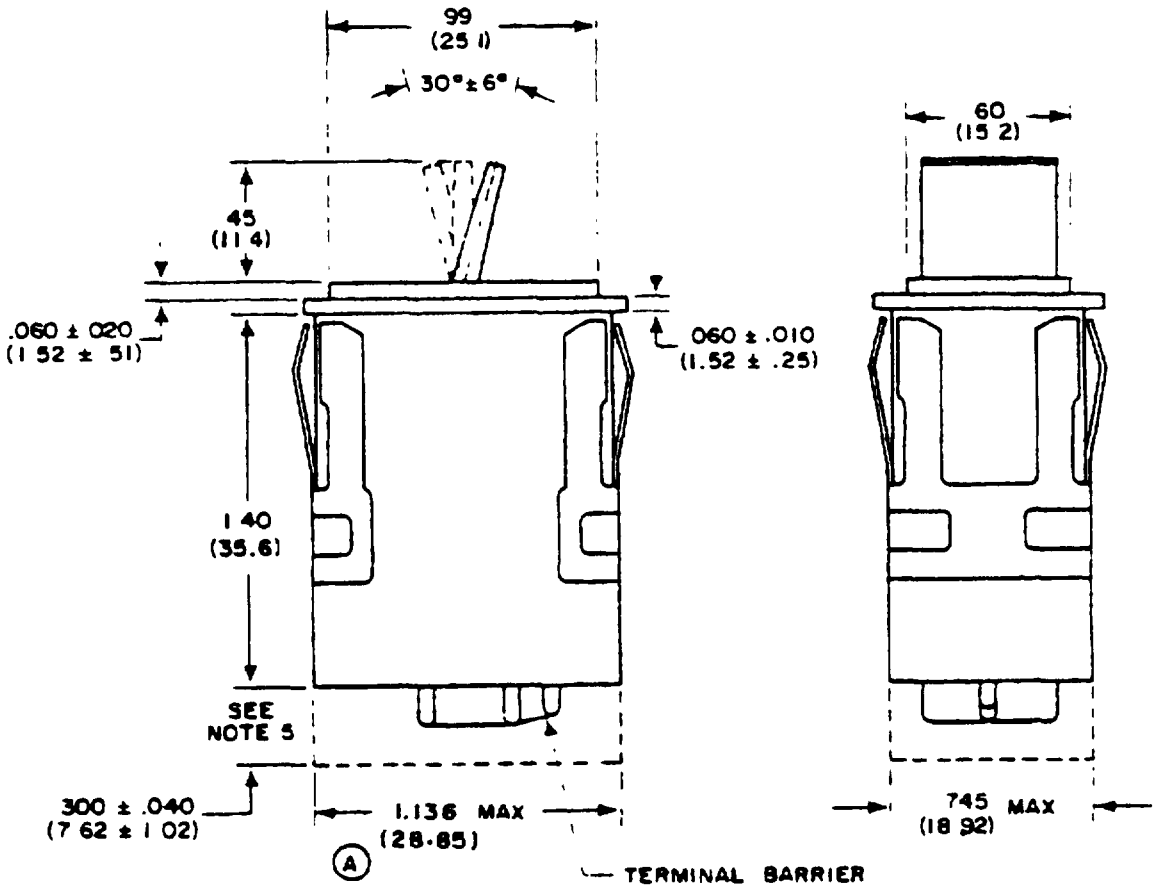
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ROCKER

FIGURE 1 Configurations and dimensions - Continued

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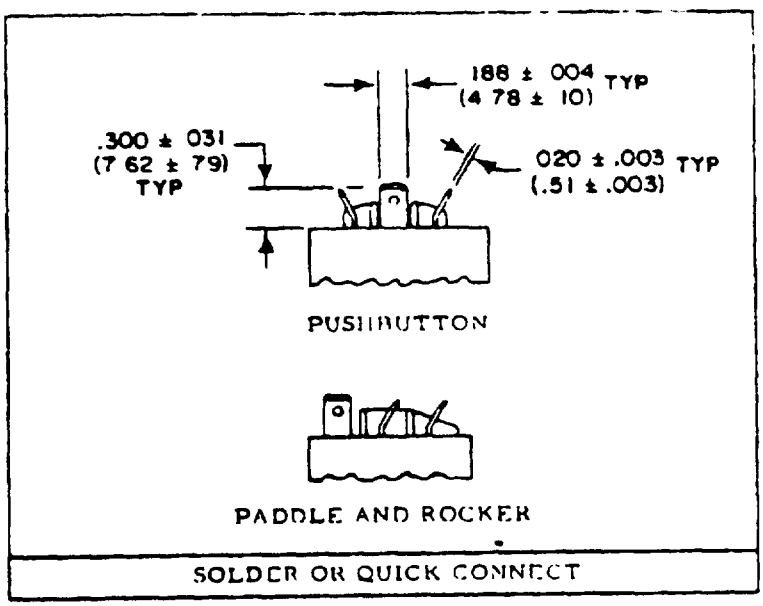


PADDLE

NOTES.
1. Dimensions are in inches.
2. Tolerances are ±.031 (79 mm) unless otherwise specified.
3. Switches shall mount on .060-.187 (1.52-4.75 mm) panel thickness.
4. The design configuration is optional within envelope dimensions shown.
5 See figures 3 and 4 for terminal size and location.
6. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
7. Metric equivalents are in parentheses.

FIGURE 1 Configurations and dimensions - Continued.

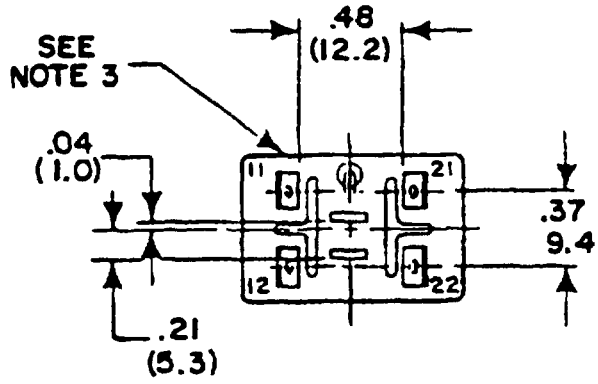
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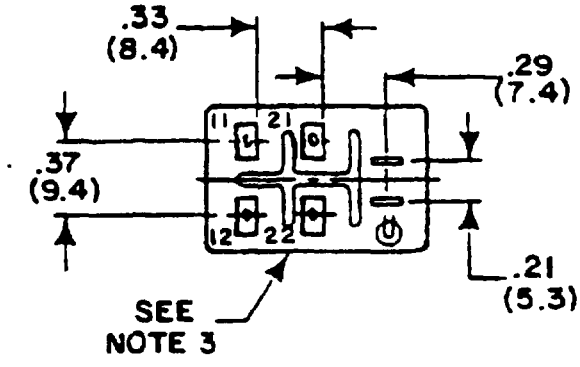
NOTES.
1. Solder hole will accept one #14 AWG stranded conductor (per NEMA publication DC-2 1976).
2. Dimensions are in inches.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Metric equivalents are in parentheses.

FIGURE 2. Terminals.

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PUSHBUTTON



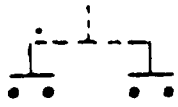
PADDLE AND ROCKER

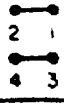
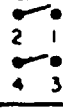
NOTES:
1. Dimensions are in inches.
2. Tolerances are ±.031 (.79 mm) unless otherwise specified.
3. Manufacturer's logo on this side of housing.
4 Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
5. Metric equivalents are in parentheses.

Ⓐ FIGURE 3. Terminal location.

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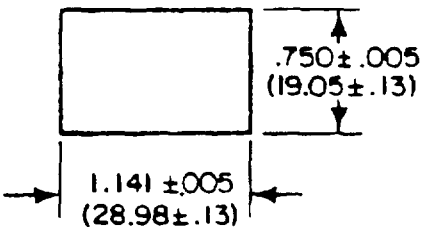
TABLE I Circuitries

PUSHBUTTON	
POLES	CIRCUITRIES
2 (FORM X)	

ROCKER AND PADDLE		
POLES		
1 (FORM A)		
2 (FORM A)		

NOTE: Circuitries are shown with the switch positioned such that the manufacturer's logo is located on this side of the switch

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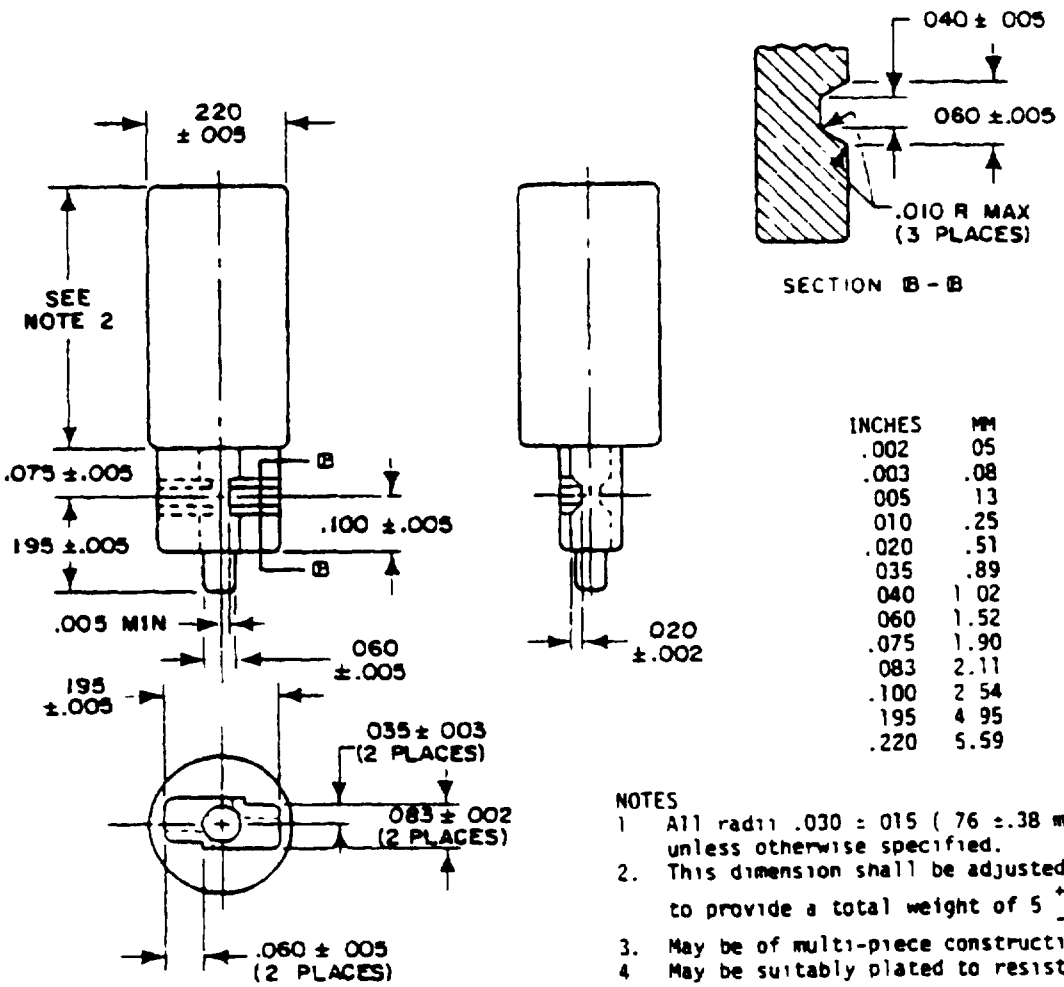


Ⓐ	STRIP MOUNTING (SEE NOTE 3)
	(1.20)(NO. OF UNITS) -.045 (30,5)(NO. OF UNITS) - 1,14

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm.
3 See application information for strip mounting information

FIGURE 4 Panel cutout.

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INCHES	MM
.002	05
.003	.08
005	13
010	.25
.020	.51
035	.89
040	1 02
060	1.52
.075	1.90
083	2.11
.100	2 54
195	4 95
.220	5.59

NOTES
1. All radii .030 ± 015 (76 ±.38 mm) unless otherwise specified.
2. This dimension shall be adjusted to provide a total weight of 5 $^{+1}_{-0}$ gram.
3. May be of multi-piece construction
4. May be suitably plated to resist corrosion
5. Dimensions are in inches.
6. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 5. T1-3/4 wedge base lamp retention plug.

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REQUIREMENTS

Dimensions and configuration See figures 1 through 5 and table .

Complete switch shall consist of

Switch housing - The switch housing shall include mounting clips

Actuation and display - Unassembled, when furnished.

Pushbutton actuation

Transmitted color (one translucent colored button is furnished when specified)
Projected color (one white translucent cap and a translucent color insert is furnished when specified)
Hidden color (one black appearing cap and a translucent color insert is furnished when specified)
Nonilluminated (one translucent colored button is furnished when specified)

Rocker actuation - Rocker actuation provides transmitted color display only The rocker operator can be provided as one piece full rocker or two piece rocker for contrasting color selection
Paddle actuation - Same features as detailed for the pushbutton actuation except paddle cover is provided as a two piece cover

Lamp - Not furnished Illuminated devices will accept T-1 3/4 subminiature wedge base incandescent lamp

Enclosure design 1 (unsealed)

Temperature characteristic 0° to +55°C

Vibration grade 2 (10-500 Hz)

Material.

Polycarbonate May be used for light display and actuator materials only, provided it meets UL 94V-2 or better

Weight See table II

TABLE II Weight

Configuration	Maximum weight
Pushbutton	31 grams
Rocker	35 grams
Paddle	35 grams

Lamp contacts Contacts for incandescent lamps shall be designed to accept T-1 3/4 wedge base lamps.

Lens The lens design shall permit application of hot stamp legends

Explosion Not applicable

Overload Inductive load ac electrical endurance samples shall be subjected to a resistive overload test at 150 percent of the inductive rating

Electrical ratings and endurance See table III

Operating characteristics See table IV

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TABLE III. Electrical ratings and endurance

Rating	Cycles
Pushbuttons	
10 amperes, 125 Vac (60 power factor) 3 amperes 28 Vdc resistive	25,000
Rockers and paddles	
15 amperes, 125 Vac (60 power factor) 10 amperes 28 Vdc resistive	25,000

TABLE IV. Operating characteristics

Configuration	Characteristic	Value
Pushbutton	Total actuator travel	170 in min
Rocker		30° -6°
Paddle		30° -6°

Contact resistance (switch)

Initial 100 ohms maximum
After electrical endurance Not applicable
Lamp contact resistance Lamp circuit resistance test plug in accordance with figure 6

Strength of actuating means

Rocker actuation - The static load shall be gradually applied to the rocker surface when that surface is parallel to the mounting surface
Paddle actuation - With the paddle in the extreme position a static load of 15 pounds shall be gradually applied perpendicular to the paddle axis and parallel to the direction of paddle travel

Thermal shock Temperature extremes shall be -40° to +65°C

Shock (specified pulse) MIL-STD-202, method 213, test condition A (50 G's)

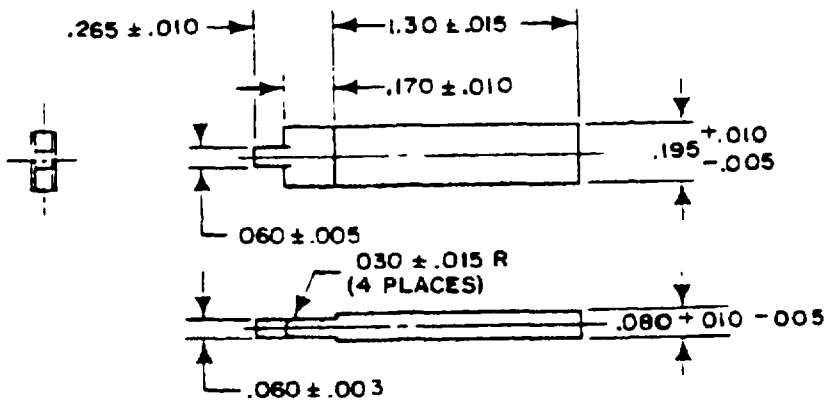
Field of view Obstruction of view by the paddle shall not be cause for failure

Color and luminance See tables V and VI

Group A inspection See table VII.

Qualification inspection See table VIII

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INCHES	MM
.003	.08
.005	.13
.010	.25
.015	.38
.030	.76
.060	1.52
.080	2.03
.170	4.32
.195	4.95
.265	6.73
1.30	33.02

NOTES.
1. Dimensions are in inches
2. Material shall be 1/2 hard brass
3. May be suitably plated to resist corrosion
4. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm

Ⓐ FIGURE 6. T1-3/4 wedge base lamp circuit resistance test plug

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(A) TABLE V Illuminated chromaticity limits 1/2/
Incandescent

Color	Chromaticity coordinates					
	Projected color		Transmitted color		Hidden color	
	"X"	"Y"	"X"	"Y"	"X"	"Y"
Blue	275	525	325	475	275	475
	275	450	250	475	225	475
	200	425	250	400	200	400
	200	525	325	400	275	350
Green	415	495	290	610	275	650
	375	450	255	525	225	595
	300	525	180	610	225	500
	355	585	180	695	375	550
Red	695	285	695	285	695	285
	710	290	710	290	710	290
	655	320	655	320	655	320
	660	340	660	340	660	340
White	550	395	475	435	510	395
	530	435	475	395	530	435
	460	435	530	435	460	435
	460	385	545	395	460	395
Yellow	580	400	580	400	580	400
	590	410	590	410	590	410
	515	450	515	450	515	450
	525	475	525	475	525	475

1/ Measured at 2200° Kelvin using one configuration type II indicator housing plus two lens of each color

2/ Illuminated chromaticity limits shall be established by the spectroradiometric method for incandescent lamps.

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Ⓐ TABLE VI Brightness - incandescent switches (pushbutton, rockers, and paddles).

Color	Brightness (foot lamberts), minimum average								
	Pushbutton, configuration type I 1/			Rockers, configuration type II 2/			Paddles, configuration type III 2/		
	Projected	Transmitted	Hidden	Projected	Transmitted	Hidden	Projected	Transmitted	Hidden
Blue	11	7	2	22	12	4	22	12	4
Green	22	11	2	36	17	4	36	17	4
Red	19	15	3	38	34	8	38	34	8
White	58	43	12	98	98	17	98	98	17
Yellow	52	62	13	108	123	20	108	123	20

1/ Tested with one lamp using one housing plus two lens of each color.

2/ Tested using one housing plus two lens of each color.

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TABLE VII Group A inspection

Inspection	AQL percent defective	
	Major	Minor
Visual and mechanical inspection	1.0 1/	4.0
Operating characteristics	1.0	---
Dielectric withstanding voltage	65	---
Contact resistance (switch)	65	---

1/ At the option of the contractor, in-process inspection may be used to meet the materials, and design and construction requirements provided they meet the acceptable quality level and all of the contractor's in-process control data on these tests are made available to the Government upon request

TABLE VIII Qualification (group submission)

Inspection	Sample	Ext of approval
Group I Visual and mechanical inspection 1/ Solderability 2/ Resistance to solder heat 2/ Contact resistance Operating characteristics	All	All
Group II Terminal strength 2/ Strength of actuating means 3/ Lamp retention Thermal shock Vibration Shock (specified pulse) Moisture resistance Dielectric withstanding voltage	(A) M22885/92-8AA11W (4 samples) M22885/92-FAA21FWW (2 samples) M22885/92-DAA21BWW (2 samples)	
Group III Salt spray (corrosion)	(A) M22885/92-8AA11W (2 samples) M22885/92-FAA21FWW (2 samples)	
Group IV Short circuit Dielectric withstanding voltage Operating characteristics	(A) M22885/92-8AA11W (2 samples) M22885/92-FAA21FWW (2 samples)	
Group VI Overload Electrical endurance Contact resistance (lamp) Dielectric withstanding voltage Operating characteristics	(A) M22885/92-8AA11W (4 samples) M22885/92-FAA21FWW (2 samples) M22885/92-DAA21BWW (2 samples)	
Group VIII Color Luminance Field of view	Chromaticity and brightness samples to be tested are as determined by the applicable tables. Two samples of each configuration are tested for field of view	

1/ Two samples of each configuration type for physical dimensions
2/ Two samples of each terminal type.
3/ Two samples of each actuator type

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Part number for pushbutton switches:

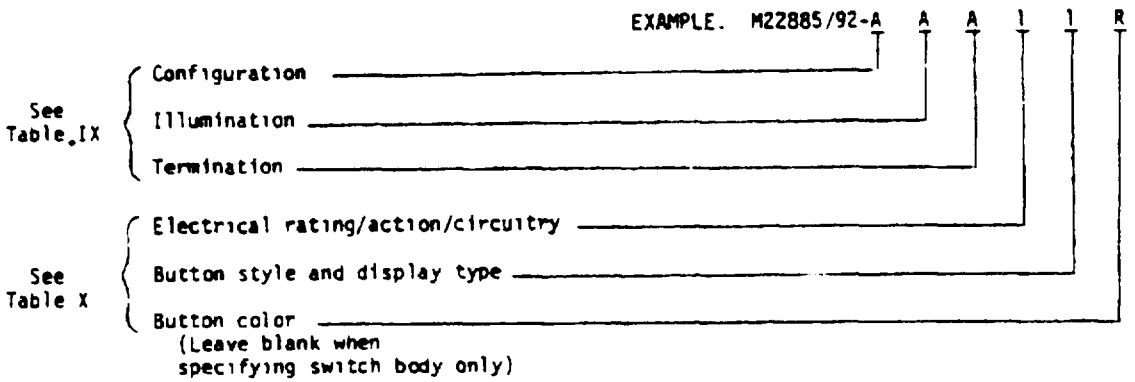


TABLE IX Configuration illumination and terminal codes for pushbutton switches

Configuration	Illumination	Termination
A - Rectangular pushbutton non-illuminated B - Rectangular pushbutton 1 lamp circuit	A - No lamps installed	A - Solder/quick connect

TABLE X. Electrical rating, action, circuitry, display and button style for pushbutton switches.

Electrical rating/action/circuitry	Button style and display type	Button color
1 - 10 Amperes, 125 Vac (60% pf) 8 Amperes, 28 Vdc resistive Alternate Action/2 Pole 2 - 10 Amperes, 125 Vac (60% pf) 8 Amperes, 28 Vdc resistive Momentary Action/2 Pole	1 - Rectangular pushbutton, transmitted color 1/ 2 - Rectangular pushbutton, projected color 3 - Rectangular pushbutton, hidden color 4 - Switch body only. No buttons, caps, or color inserts supplied (Procure separately from source listed on QPL)	R - Red Y - Yellow G - Green B - Blue W - White K - Black** L - Gray** **For non-illuminated configuration type only

1/ Use transmitted color for nonilluminated switches

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Part number for rocker and paddle switches

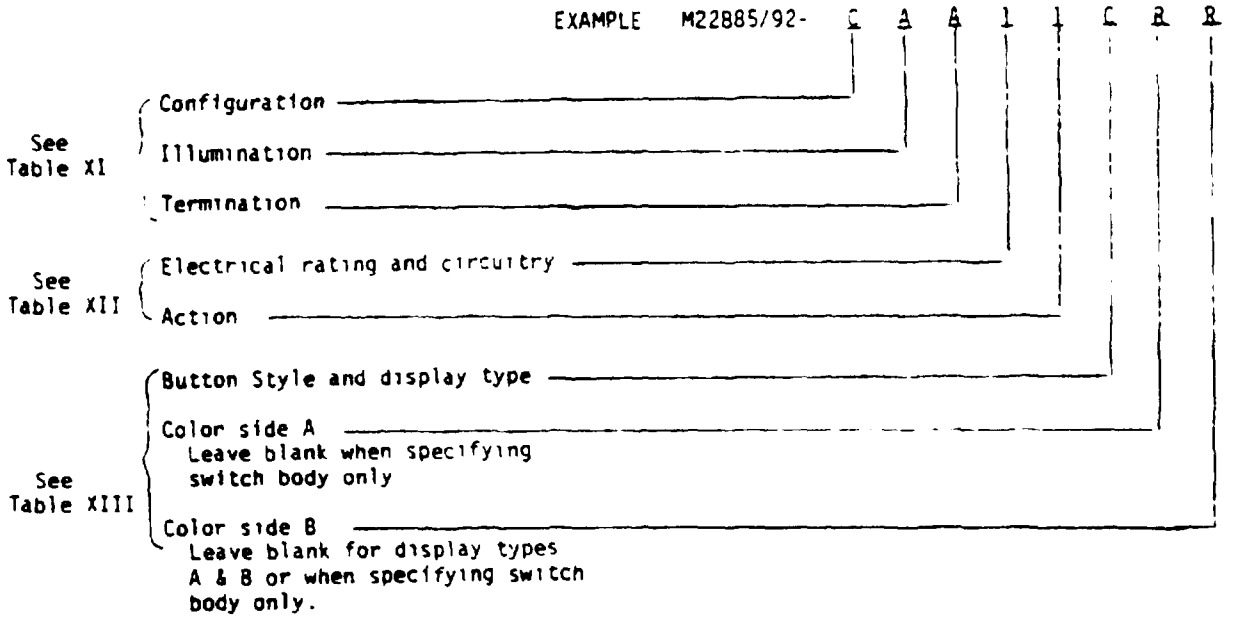
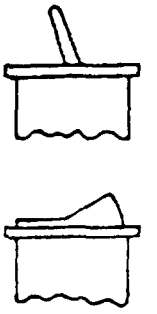
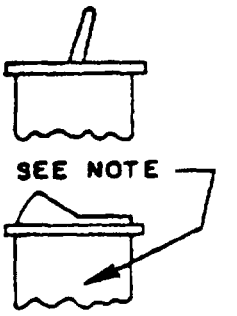


TABLE XI Configuration, illumination and terminal codes for rocker and paddle switches

Configuration	Illumination	Termination
C - Rocker, nonilluminated	A - No lamp installed	A - Solder/quick connect
D - Rocker, 1 lamp circuit		
E - Paddle, nonilluminated		
F - Paddle, 1 lamp circuit		

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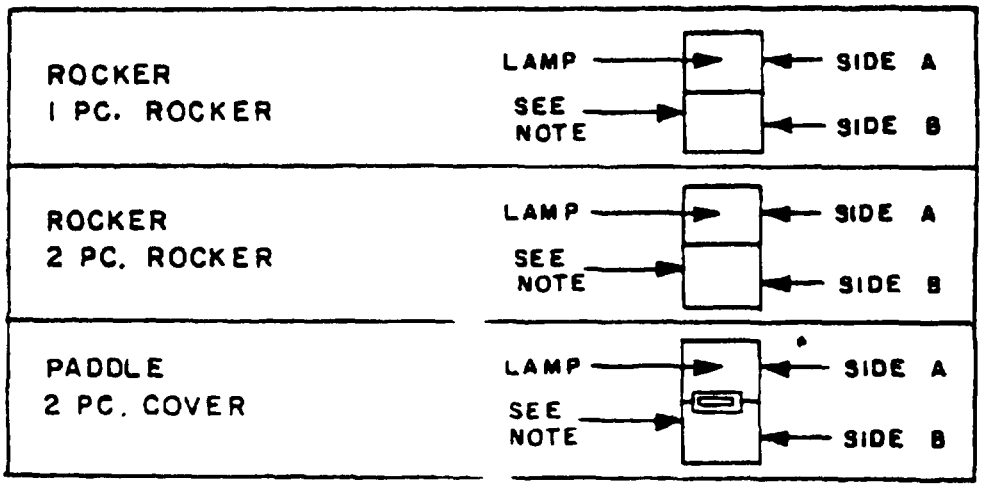
TABLE XII Electrical rating, circuitry and action codes for rocker and paddle switches

Electrical rating/circuitry	ACTION CODE		
1 - 15 Amperes, 125 Vac (60% pf) 10 Amperes, 28 Vdc resistive 1 Pole 2 - 15 Amperes, 125 Vac (60% pf) 10 Amperes, 28 Vdc resistive 2 Pole			
	I	MAINTAINED	MAINTAINED

NOTE: Actions are shown with the switch positioned such that the manufacturer's logo is located on this side of the switch.

TABLE XIII. Button style, display type and color for rocker and paddle switches

Button style/display type	Color	
A - Nonilluminated, 1 piece rocker (one color only) B - Transmitted color, 1 piece rocker (one color only) C - Nonilluminated, 2 piece rocker D - Transmitted color, 2 piece rocker E - Nonilluminated, 2 piece paddle F - Transmitted color, 2 piece paddle G - Projected color, 2 piece paddle H - Hidden color, 2 piece paddle J - Switch body only - no covers or inserts supplied Procure from source listed on QPL	Side A R - Red Y - Yellow G - Green B - Blue W - White K - Black* L - Gray*	Side B R - Red Y - Yellow G - Green B - Blue W - White K - Black* L - Gray*
	*For nonilluminated configuration type.	



NOTE. Button styles are shown as viewed from the front of the panel and positioned so that the manufacturer's logo is on the side of the housing

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APPLICATION INFORMATION

RECOMMENDED STRIP-MOUNTING ORIENTATION

Pushbuttons. "Panel holding" mounting clips are on the long sides. Only horizontal strip mounting with the long sides adjacent the panel edges is recommended. (No more than three units can be strip mounted with the long bezel sides touching.)

Rockers and paddles. "Panel holding" mounting clips are located on the short sides. This permits horizontal strip-mounting with the long sides of the bezel touching. (No more than three rocker or paddle switches can be strip-mounted with the short sides of the bezel touching.)

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